

Pranshu Patel . Jarvis Consulting

Aspiring Mainframe Developer with a strong foundation in data engineering, SQL, Linux, and enterprise application development. Experienced in working with Python, SQL, Git, relational databases, and data processing workflows, with growing hands-on knowledge of COBOL, JCL, DB2, ISPF, z/OS commands, and batch-processing concepts. Developed mainframe-style COBOL applications involving file handling, indexed files, copybooks, record layouts, CRUD operations, sorting, reporting, and structured business logic. Strong understanding of data quality, automation, backend processing, and relational database systems, with a keen interest in contributing to COBOL, DB2, JCL, and enterprise batch application development in large-scale banking and business environments.

Skills

Proficient: SQL, Linux/Bash, Git, Relational Databases, Data Analysis

Competent: COBOL, DB2, JCL, ISPF, z/OS Commands, Batch Processing, File Handling, Python

Familiar: Mainframe, Copybooks, Record Layouts, Indexed Files, RDBMS, Agile/Scrum, Docker, Power BI, Pandas, PySpark, Cloud Computing, Tableau, SAS, Excel Automation

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_PranshuPatel

COBOL Student Registration System [GitHub]: Developed a mainframe-style Student Registration System using COBOL in OpenCobolIDE/GnuCOBOL to simulate enterprise batch and transaction-processing logic. Built a menu-driven application to generate student files, insert records, update records, delete records, query students by ID or inclusion date, and produce course-wise reports. Applied structured COBOL programming concepts including divisions, paragraphs, working-storage variables, copybooks, record layouts, file handling, indexed file processing, sorting, CRUD operations, and report generation. This project strengthened my understanding of COBOL business logic, batch-style processing, file organization, and core concepts used in mainframe application development.

Linux Cluster Resource Monitoring App [GitHub]: Developed a Linux cluster monitoring application using CentOS, Bash, Docker, PostgreSQL, and crontab to collect, store, and analyze system resource metrics, enabling automated data analytics, system health tracking, and performance reporting across multiple nodes.

RDBMS and Advanced SQL Analytics [GitHub]: Designed and executed advanced SQL queries on PostgreSQL to analyze system and business datasets, utilizing joins, subqueries, window functions, indexing, and aggregations to optimize data retrieval, support analytics use cases, and generate meaningful insights from relational databases.

Python Data Analytics [GitHub]: Processed and analyzed a large-scale dataset of over 1M records by importing relational SQL data into Jupyter Notebook using SQLAlchemy and psycopg2, performing data transformation and analysis with NumPy and Pandas, and generating insight-driven visualizations with Matplotlib, Seaborn, and Squarify to support data-driven decision-making.

Java Grep Application [GitHub]: Designed and implemented a Unix-style grep application in Core Java 8 that recursively scans directories, performs regex-based pattern matching, and writes filtered results to an output file using a clean interface-driven architecture and modular workflow design. Leveraged Java Stream and Lambda APIs to adopt functional programming practices aligned with Spark/Scala big data systems, integrated structured logging using SLF4J + Log4j, packaged the application as an executable Uber/Fat JAR via Maven Shade Plugin, explored JVM memory tuning (-Xms, -Xmx) to analyze performance and OutOfMemoryError scenarios, and containerized the application using Docker (OpenJDK Alpine base image) for environment-independent deployment and distribution through Docker Hub.

PySpark Data Analysis [GitHub]: Implemented a PySpark-based data analytics project across Databricks and Zeppelin to evaluate distributed data processing in notebook environments. In Databricks, loaded retail transaction data from PostgreSQL using JDBC and built end-to-end analytics with PySpark DataFrames, including monthly sales, placed vs. canceled orders, sales growth, active users, new vs. existing users, and RFM-based customer segmentation. In Zeppelin, analyzed World Development Indicators data using PySpark and Hive-backed tables to strengthen Spark transformation, filtering, aggregation, and join skills. Applied structured DataFrame APIs, data wrangling, and business-focused analysis to generate scalable insights across customer behavior and sales trends.

ETL pipeline for Financial Data [GitHub]: Built an end-to-end Financial Fraud Analytics ETL pipeline on Azure Databricks using a medallion architecture (Bronze, Silver, Gold) to ingest, clean, enrich, and analyze multi-source financial transaction data. The project integrated Azure SQL Database, Azure Data Lake Storage Gen2, PySpark, JDBC,

Databricks dashboards, and Databricks Jobs to process transactions, card data, user data, merchant category codes, and fraud labels into analytics-ready datasets. I designed gold-layer tables to answer key fraud business questions such as fraud trends over time, high-risk users, merchant category fraud rates, daily fraud losses, and behavior changes before and after fraudulent events, and then built an interactive dashboard with filters and job orchestration to automate the full pipeline from raw ingestion to reporting refresh.

DLT pipeline for Alpha-Vantage Stock Data [GitHub]: Designed and implemented an end-to-end stock market analytics pipeline in Databricks using Delta Live Tables (DLT), PySpark, and Alpha Vantage API, processing daily stock data for AAPL, MSFT, GOOGL, and TSLA through a medallion architecture. Built bronze, silver, and gold layers to ingest raw API feeds, enforce data quality checks and business rules, standardize and deduplicate records, and produce analytical tables for 7, 30, and 90-day price and volume trend analysis. Integrated the gold layer with a Databricks dashboard for business-friendly visualization and automated the entire workflow using Databricks Jobs, including ingestion, pipeline refresh, and dashboard update, strengthening my practical experience in DLT-based ETL design, financial data engineering, and workflow orchestration.

Highlighted Projects

E-Commerce Sales Data Pipeline [GitHub]: Engineered an end-to-end ETL pipeline using Python, SQL, and Pandas to process over 100k e-commerce transactions, performing data cleansing and transformation to support Power BI dashboards and improve analytics-driven business insights.

Retail Inventory Optimization System: Designed a SQL-driven inventory monitoring system analyzing over 200k product records to identify reorder points and stock discrepancies, automating KPI reporting with Python and Power BI to reduce stockouts and improve inventory planning efficiency.

Professional Experiences

Data Engineer / Mainframe Developer Trainee, Jarvis Consulting Group (January 2026-present): Worked on enterprise application development and data engineering projects using COBOL, SQL, Linux, Java, Python, Docker, Git, and relational databases. Developed mainframe-style COBOL applications involving file handling, indexed files, record layouts, copybooks, menu-driven workflows, CRUD operations, sorting, and reporting. Strengthened understanding of DB2/SQL, JCL, ISPF, z/OS commands, batch processing, debugging, technical documentation, and Agile collaboration while preparing for client-facing Mainframe Developer responsibilities.

Data Engineer Intern, Techomax Solutions (February 2023-April 2023): Validated and cleaned transactional datasets using Python, SQL, and SAS, performed exploratory data analysis, and built BI dashboards to enhance data reliability and analytical reporting for client-facing deliverables.

Data Engineer Intern, Webburner India (June 2022-July 2022): Collected, cleansed, and structured CRM and survey data using Python, SQL, and Excel while automating ETL workflows and developing dashboards to support marketing analytics and performance tracking.

Education

Northeastern University, Toronto (January 2024-June 2025), Master of Professional Studies, Informatics (Cloud Computing)

Gujarat Technological University, India (July 2019-May 2023), Bachelor of Engineering, Computer Science & Engineering

Miscellaneous

- COBOL Essential Training - LinkedIn
- IBM Badge: Introduction to z/OS Commands and Panels on IBM Z
- Career Essentials in Data Analysis Microsoft & LinkedIn
- Foundations & Career Skills in Data Analytics LinkedIn
- Database Management Systems & Data Science with Python Scaler
- Continuous learning in data engineering and analytics
- Technical documentation and project reporting